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**Dirty Air and Green Investments: The Impact of Pollution Information on
Portfolio Allocations**

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1 Big picture

- **Question.** Does access to local pollution information cause retail investors to shift their portfolios away from brown stocks and toward greener allocations?
- **Setting.** The paper studies India, where Continuous Ambient Air Quality Monitoring Stations (CAAQMSs) were rolled out across locations over time. These stations made real-time local pollution information publicly available through online and mobile channels.
- **Key treatment idea.** Investors located within 20 kilometers of a new CAAQMS are treated because the station reading was communicated as locally relevant. Investors in the 40–60 kilometer donut ring around the same station are the main control group.
- **Main data.** The paper uses the universe of National Stock Exchange of India retail trading records, allowing the authors to construct portfolio brown shares for about 19 million retail investors over 2004–2020.
- **Outcome.** The main dependent variable is the share of brown stocks in retail investors’ portfolios at the PIN-code-month level. Brown industries are high-emission sectors, and green stocks are the complement.
- **Main result.** After a nearby CAAQMS appears, the relation between local pollution and brown portfolio share becomes significantly more negative for treated investors. Before the station appears, this relation is approximately flat.
- **Economic magnitude.** A one-standard-deviation worsening in within-station air quality leads treated investors to reduce brown share by about 0.5 percentage points, or roughly 1.25 percent of the mean.
- **Mechanism.** The effect is driven by selling brown stocks, not by buying green stocks. It is stronger on salient alert days when AQI crosses from Moderate to Poor, and it is strongest among mobile traders, young investors, and female investors.
- **Interpretation.** The paper argues that information salience changes investor tastes or attention rather than providing new expected-return information. Indeed, brown stocks sold on high-pollution days later earn positive excess returns, implying a cost to the divestment behavior.
- **Takeaway.** Retail ESG behavior depends not only on preferences but also on the visibility and salience of environmental problems. Information infrastructure can shift household portfolios.

2 Incremental contribution of the paper

- **From ESG preferences to information salience.** Prior ESG portfolio studies show that investors’ environmental preferences and personal experiences matter. This paper adds that the same underlying environmental condition may not affect portfolios unless information is made salient and easily accessible.
- **Clean policy shock to salience.** The rollout of CAAQMSs is not an environmental cleanup or a change in firms’ fundamentals; it is a change in the public availability and timeliness of pollution information. This lets the paper separate salience from actual pollution changes.
- **Granular household trading data.** The paper uses trading records for roughly 19 million Indian retail investors, geocoded to PIN Codes, which is unusually rich for studying household sustainable-investment behavior.
- **Triple-difference design.** The central object is not the level of brown holdings; it is the change in the sensitivity of brown holdings to local pollution after public information arrives, comparing treated and donut-control PIN Codes.
- **Mechanism evidence from trade direction.** The paper distinguishes brown divestment from green purchases using order imbalance. This shows the main response comes from selling brown firms, not from new purchases of green firms.

- **High-frequency salience test.** The regression discontinuity around the AQI 200 Moderate-to-Poor threshold links the portfolio response to an explicit public warning threshold, strengthening the salience interpretation.
- **Investor heterogeneity.** The paper shows larger effects among mobile traders, young investors, and female investors, groups plausibly more exposed to pollution data or more responsive to environmental concerns.
- **Evidence of nonpecuniary motives.** Counterfactual return evidence suggests that brown-stock sales would have earned positive excess returns, implying that the behavior likely reflects preferences, salience, or warm-glow utility rather than rational return chasing.
- **Policy relevance.** The paper provides direct evidence that information dissemination can shift retail capital allocations, which is important because retail investors can either complement or offset institutional ESG tilts.

3 Data and measurement

3.1 Investor-level trading and portfolio construction

- Trading records come from the National Stock Exchange of India.
- Trades can be linked to anonymized permanent account numbers and to investor PIN Codes.
- The paper aggregates to the PIN-code-month level, constructing the average brown share held by local retail investors.
- Brown share is calculated for all investors and by subgroups: female, male, young, middle-aged, and old investors.
- Order-imbalance measures use daily buys and sells to distinguish brown divestment from green purchase behavior.

Interpretation: The data are powerful because they connect geography, local pollution information, and actual portfolio holdings/trades rather than survey attitudes.

3.2 Pollution data and CAAQMS rollout

- CAAQMSs provide real-time automated pollution information and AQI readings.
- The first station appeared in 2007, with a broad rollout during the 2010s.
- The paper defines treatment using the 20-kilometer public relevance radius around stations.
- Control PIN Codes are in a 40–60 kilometer donut ring around the same station.
- Satellite Aerosol Optical Depth (AOD) data provide a consistent pollution proxy before and after station installation.

Interpretation: The key empirical advantage is that pre-rollout pollution can still be measured using satellite data, while public salience changes only when CAAQMS readings become available.

3.3 Brown and green stocks

- Brown stocks are classified using high-emission industries, based on IPCC categories and Indian firm data.
- Green stocks are defined as publicly traded stocks outside the brown classification.

- The main outcome is brown share, not absolute holding amount, because ESG-augmented portfolio theory emphasizes the weighted ESG composition of portfolios.
- Robustness checks use alternative brown definitions, including high-emitter firms from Thomson Reuters' global greenhouse gas emitters list.

Interpretation: The paper intentionally studies portfolio composition rather than only transaction flows because ESG preferences are naturally portfolio-share objects.

3.4 Controls and heterogeneity variables

- Local trading controls include number of investors and turnover.
- Weather controls include rainfall and temperature.
- Heterogeneity variables include trading technology, age, and gender.
- Additional robustness examines whether results are driven by large cities, alternative treatment radii, or other stock characteristics.

Interpretation: The controls address the concern that changes in brown share may reflect local economic conditions, weather, or changes in trading intensity rather than information salience.

4 Models and empirical specifications

4.1 Triple-difference brown-share specification

The main estimating equation is a triple interaction model:

$$BrownShare_{p(m),t} = \beta Treat_p \times Pollution_{p,t} \times Post_{m,t} + \sum_k \alpha_k OtherInteractions + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}.$$

- p is a PIN Code, m is a monitoring-station area, and t is time.
- $Treat_p$ equals one for PIN Codes within 20 km of a station.
- $Post_{m,t}$ equals one after the station appears.
- γ_p are PIN Code fixed effects.
- $\lambda_{m,t}$ are station-area-by-time fixed effects.
- The coefficient β is the change in the brown-share-pollution gradient for treated investors after the monitoring station appears.

Interpretation: The specification asks whether pollution becomes more relevant for portfolio allocation only after pollution information becomes publicly available nearby.

4.2 Event-study specification

The event-study version replaces $Post_{m,t}$ with quarter-relative-to-rollout indicators:

$$BrownShare_{p(m),t} = \sum_{i=-6}^6 \beta_i Treat_p \times Pollution_{p,t} \times D(t \in i)_{m,t} + \sum_k \alpha_k OtherInteractions + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}.$$

Interpretation: The event study tests for pretrends and shows when the brown-share-pollution gradient changes relative to station rollout.

4.3 Order-imbalance specification

The paper studies whether the response comes from selling brown stocks or buying green stocks:

$$OI_{p(m),t}^{HighPollution} / OI_{p(m),t}^{LowPollution} = \beta Treat_p \times Post_{m,t} + X'_{p,t} \theta + \gamma_p + \lambda_{m,t} + \varepsilon_{p,t}.$$

Interpretation: This specification uses daily directional trading to identify whether investors rebalance by selling brown stocks, buying green stocks, or both.

4.4 Regression discontinuity at the AQI warning threshold

The paper examines the AQI cutoff of 200, where public messaging moves from Moderate to Poor and amber alerts begin:

$$BrownShare_{p,d} = f(AQI_{p,d} - 200) + \tau \mathbf{1}\{AQI_{p,d} \geq 200\} + controls + \varepsilon_{p,d}.$$

Interpretation: This design links investment behavior to a high-salience public warning threshold rather than to the gradual level of pollution alone.

5 Identification and empirical design

5.1 Core identifying variation

- Treatment is the appearance of a nearby real-time pollution station.
- The treated group is within 20 km, while the control group is a 40–60 km donut ring around the same station.
- Station-area-by-time fixed effects compare treated and control PIN Codes around the same station at the same time.
- PIN Code fixed effects absorb time-invariant local portfolio preferences and demographics.

Interpretation: The core identification compares nearby investors whose station-relevant information salience changes with slightly farther investors exposed to similar broad regional shocks but less relevant station information.

5.2 Why actual pollution changes are not the main driver

- The paper examines pollution levels around station installation.
- There is no discrete change in satellite AOD pollution when a CAAQMS appears.
- The portfolio response appears only after public monitoring information becomes available.

Interpretation: This supports the salience interpretation: investors react to information availability, not to a coincident physical change in air quality.

5.3 Why the donut-control design matters

- Donut controls are close enough to share similar regional trends, weather, and economic shocks.
- They are far enough that the station’s reported air quality is less directly relevant.
- The control group shows no comparable post-rollout change in the brown-share-pollution gradient.

Interpretation: The design is not just treated versus India-wide controls; it is local treated versus local-but-less-relevant controls.

5.4 Robustness and limitations

- Robustness checks use stacked difference-in-differences, alternative treatment radii, alternative brown definitions, exclusion of large cities, and other stock-characteristic portfolios.
- The paper cannot fully separate investor tastes from mistaken beliefs about expected returns.
- Heterogeneity by age, gender, and technology is suggestive because these investor attributes are not randomly assigned.
- The retail investor response may not generalize to institutional investors, who show no similar effect.

Interpretation: The evidence is strongest on the causal role of pollution-information salience for retail portfolio allocation. The exact preference-versus-belief channel remains an open question.

6 Section-by-section study guide

- **Introduction.** Explains why retail ESG behavior matters and previews the CAAQMS natural experiment.
- **Institutional background.** Describes India’s pollution monitoring system and the transition from manual, stale data to real-time CAAQMS data.
- **Data.** Defines retail portfolios, brown stocks, pollution, treatment/control PIN Codes, weather controls, and summary statistics.
- **Empirical strategy.** Develops the triple-difference and event-study designs.
- **Results.** Presents visual evidence, baseline regressions, robustness, order imbalance, RD salience tests, and heterogeneity.
- **Conclusion.** Interprets findings as salience-driven retail portfolio reallocations with possible foregone return costs.

7 Tables and figures

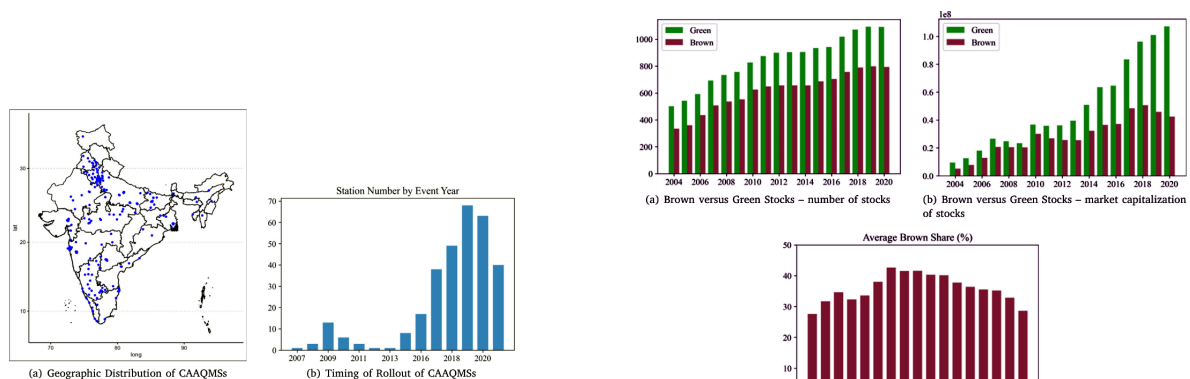
7.1 Figures 1–2: CAAQMS rollout and green/brown stock universe

Read:

- Figure 1 shows the geographic rollout of Continuous Ambient Air Quality Monitoring Stations and the timing of station introduction.
- Figure 2 shows the number, market capitalization, and market share of green and brown stocks over time.
- The station rollout is staggered, creating treatment timing variation.
- Brown firms remain a substantial part of the investable public equity market, making portfolio brown share economically meaningful.

Figure 1 is the source of treatment variation. The stations are not all installed at once; their staggered rollout allows the paper to compare pre- and post-information periods across local station areas. Figure 2 establishes that brown and green portfolios are feasible investment sets over the entire sample. The brown share is not mechanically disappearing from the market, so later shifts in retail brown holdings can be interpreted as investor reallocation rather than simple market composition.

The experiment changes information access, not the stock universe. Investors can react because there is a large and persistent set of brown and green stocks to reallocate across.



(a) Figure 1: Geographic distribution and rollout of CAAQMSs.

(b) Figure 2: Distribution of green and brown stocks over time.

CAAQMS rollout and the green/brown equity universe.

7.2 Table 1 and Figure 3: Summary statistics and no pollution jump around station installation

Read:

- Table 1 reports the main portfolio, pollution, trading, and weather variables at the PIN-code-month level.
- Average brown share is about 41.9 percent; women and young investors have slightly lower brown shares than men and older investors.
- The pollution measure is AOD, used because it is available consistently before and after CAAQMS rollout.
- Figure 3 shows that pollution levels do not shift discretely around CAAQMS installation.

Table 1 establishes the economic scale of the dependent variable. A half-percentage-point decline in brown share is meaningful relative to a mean of roughly 42 percent. Figure 3 is an identification check: if the

arrival of stations coincided with a physical pollution shock, the paper would conflate salience with actual environmental change. The near-zero pattern in Figure 3 supports the claim that the station rollout changed information access rather than local pollution.

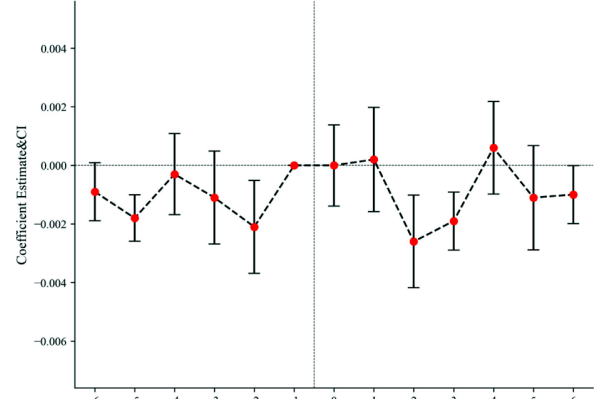
The paper's central effect is about information salience. Table 1 says the portfolio outcome has meaningful variation; Figure 3 says the treatment is not just a pollution shock.

Table 1
Summary statistics.

	mean	sd	min	p5	p10	p25	p50	p75	p90	p95	max
Brown Share (%)	41.876	9.343	0.000	28.040	32.990	37.670	41.840	45.840	51.190	56.010	100.000
Brown Share Female (%)	41.194	19.085	0.000	4.280	18.080	32.980	41.160	48.400	61.580	76.400	100.000
Brown Share Male (%)	41.935	9.644	0.000	27.640	32.340	37.600	41.900	46.010	51.540	56.580	100.000
Brown Share Young (%)	40.414	17.615	0.000	9.500	20.720	32.030	39.970	47.790	59.270	70.760	100.000
Brown Share MidAge (%)	41.945	11.481	0.000	24.440	30.750	37.180	41.910	46.490	53.100	59.250	100.000
Brown Share Old (%)	42.839	16.066	0.000	15.100	26.390	36.540	42.490	48.450	59.390	70.290	100.000
Pollution AQI	0.543	0.217	0.001	0.264	0.305	0.367	0.505	0.656	0.843	0.954	1.480
No. of Investors (Log)	3.861	1.805	0.693	1.099	1.609	2.485	3.638	4.970	6.541	7.288	10.384
Turnover (Log)	16.846	2.891	0.140	11.407	12.925	15.256	17.195	18.801	20.265	21.628	24.848
Rainfall	3.171	5.322	0.000	0.000	0.000	0.000	0.040	4.178	9.020	13.140	109.590
Temperature	25.764	4.916	3.290	15.750	18.280	23.410	26.620	29.270	31.230	32.500	36.510

Note: This table reports the summary statistics of the key variable used for the empirical analysis in this paper. The summary statistics are for the universe of Indian retail investors who trade on the National Stock Exchange, aggregated at the PIN-Code-month level. We report the average share of brown stocks for retail investors overall, and also disaggregate by gender and age cohort.

(a) Table 1: Summary statistics.



(b) Figure 3: Pollution levels around CAAQMS installation.

Summary statistics and pollution balance around treatment.

7.3 Table 2 and Figure 4: Baseline triple-difference evidence and visual binned scatter

Read:

- Table 2 estimates the main triple-difference effect of air quality information via CAAQMSs on the brown-share-pollution gradient.
- The preferred specification in Column 8 gives a negative and significant coefficient on Pollution x Treated x Post.
- Figure 4 shows that, after the station appears, treated PIN Codes display a strongly negative relation between pollution and brown share; controls do not.
- Pre-treatment panels show no comparable negative relation.

Table 2 is the main regression table. The coefficient of interest is not the level of brown share; it is the change in the sensitivity of brown share to pollution after information becomes available. Figure 4 gives the same idea visually. In the treated-post panel, high local pollution is associated with lower brown holdings, while the other three cells do not show the same pattern. This matches the triple-difference logic: the effect exists only in the treated group after information appears.

Real-time pollution information makes pollution relevant for portfolio choice. Investors reduce brown holdings specifically when pollution is locally bad and information is salient.

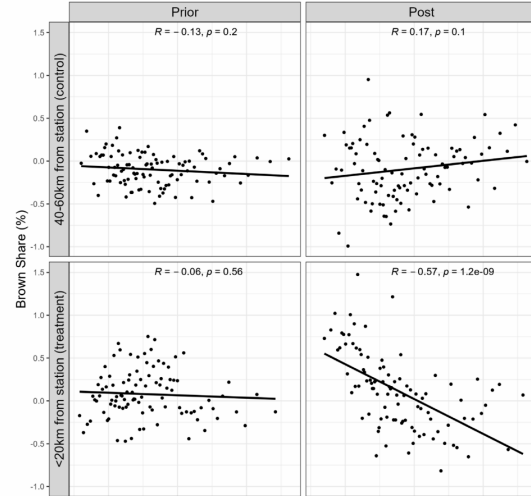
Table 2
The impact of air quality information via CAAQMS on the brown-share-pollution gradient.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. Var. = Brown Share (%)								
Pollution × Treated × Post	-1.5319** (0.7715)	-2.1017** (0.8397)	-1.7251* (0.9762)	-2.5007*** (0.9617)	-1.3634* (0.7675)	-2.0087** (0.8365)	-1.5687 (0.9739)	-2.4050** (0.9607)
Pollution × Treated	0.7596 (0.3914)	1.1181*** (0.4283)	0.9427 (0.4958)	1.2964*** (0.4835)	0.6626* (0.3893)	1.0311** (0.4270)	0.8627 (0.4945)	1.2234** (0.4826)
Pollution × Post	0.2348 (0.5418)	0.8557 (0.6750)	0.7932 (0.6974)	2.0942 (1.0528)	-0.0026 (0.5387)	0.7014 (0.6729)	0.7592 (0.6951)	1.8593 (1.6451)
Treat × Post	0.9842*** (0.2711)	0.6367** (0.2733)	0.5494 (0.3393)	0.6440** (0.3027)	0.5404** (0.2670)	0.3639 (0.2764)	0.3871 (0.3367)	0.3974 (0.3995)
Pollution	-0.2475 (0.2557)	-0.6442* (0.3094)	-0.4243 (0.7499)	-1.0167 (0.7642)	-0.1799 (0.2520)	-0.5311 (0.3885)	-0.4984 (0.7272)	-0.9371 (0.7607)
Post	-0.2702 (0.1658)	-0.0112 (0.1946)	-0.0175 (0.2513)	-0.1455 (0.1645)	0.0196 (0.1942)	-0.0172 (0.2209)	-0.0172 (0.2209)	-0.0172 (0.2209)
No. of Investors (Log)				-2.1658*** (0.1905)	-1.6582*** (0.1987)	-1.3128*** (0.2034)	-1.5185*** (0.2034)	-1.5185*** (0.2034)
Turnover (Log)				-0.0071*** (0.0046)	-0.1069*** (0.0467)	-0.0925*** (0.0413)	-0.1023*** (0.0409)	-0.1023*** (0.0409)
Rainfall				-0.0024 (0.0049)	-0.0079 (0.0054)	-0.0011 (0.0084)	-0.0114 (0.0101)	-0.0114 (0.0101)
Temperature				0.0100 (0.0066)	-0.0561* (0.0112)	-0.1447*** (0.0372)	-0.0566 (0.0734)	-0.0566 (0.0734)
Pincode	Y	Y	Y	Y	Y	Y	Y	Y
Year-month	Y	Y	Y	Y	Y	Y	Y	Y
State × Year-month		Y			Y		Y	
District × Year-month			Y	Y				Y
Rainfall × Year-month								
Observations	499,036	499,036	488,681	498,310	499,036	499,036	488,681	498,310
R-squared	0.537	0.546	0.599	0.560	0.548	0.600	0.561	0.561

Note: This table shows the elasticity of the share of brown stocks held in investors' portfolios with respect to information on pollution following the installation of Continuous Ambient Air Quality Monitoring Stations. Specifically we report the coefficient β from the following specification:

$$\text{Brown Share}_{i,t} = \beta \times \text{Treated}_i \times \text{Pollution}_{i,t} \times \text{Post}_{i,t} + \sum_{j=1}^n \alpha_j \times \text{Other Interactions}$$

(a) Table 2: CAAQMS impact on the brown-share-pollution gradient.



(b) Figure 4: Brown share and pollution by treated/control and pre/post.

Baseline regression and visual evidence.

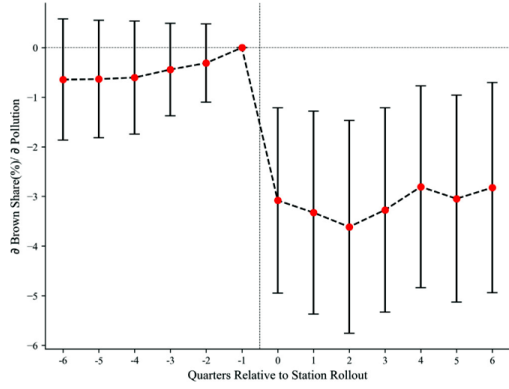
7.4 Figures 5–6: Event-study timing and treated-versus-control decomposition

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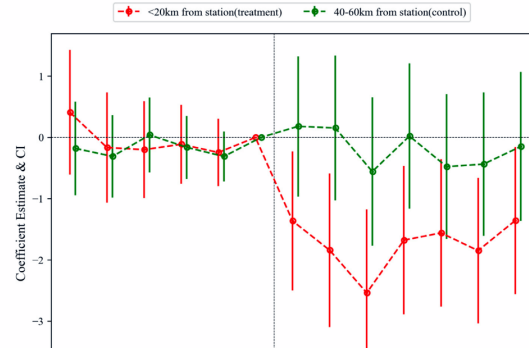
- Figure 5 plots the event-study coefficients for the treated-versus-control brown-share-pollution gradient around station rollout.
- The pre-period coefficients are near zero; the post-period coefficients turn sharply negative.
- Figure 6 separates treated and control investors. The negative post-rollout effect appears among treated investors, not donut-control investors.
- Both figures support the interpretation that the effect begins when local pollution data become available.

Figure 5 is the dynamic version of Table 2: it shows no pretrend and an immediate post-treatment break. Figure 6 addresses a lower-order concern in Table 2 by allowing treated and control groups to have different time paths. The red treated line falls after rollout; the green control line remains close to zero. This makes the identification story more credible because the effect is not mechanically created by pooling treated and control PIN Codes.

The timing is consistent with salience. Investors are not generally drifting toward greener portfolios before the station; they begin reacting when local pollution information becomes visible.



(a) Figure 5: Sensitivity of brown share to pollution.



(b) Figure 6: Treated versus control event study.

Dynamic event-study evidence.

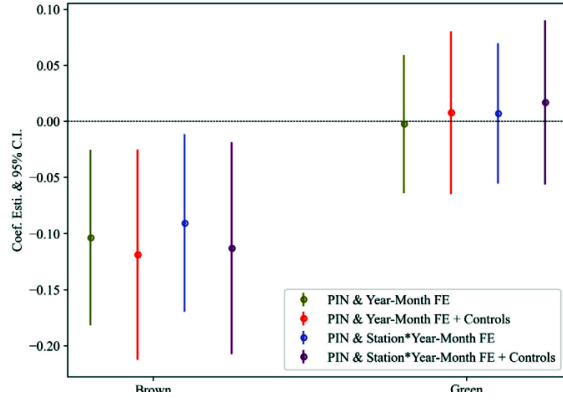
7.5 Figures 7–8 and Table 3: Trade direction and AQI warning-threshold salience

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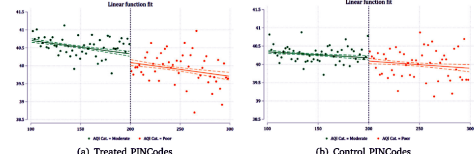
- Figure 7 shows that the portfolio response is driven by brown-stock selling rather than green-stock buying.
- Figure 8 shows regression discontinuity plots around the AQI 200 threshold, where public warnings move from Moderate to Poor.
- Table 3 reports RD estimates; treated PIN Codes show significant negative discontinuities in brown share above the warning threshold, while controls do not.
- This evidence links the portfolio response to a salient public warning cutoff rather than only to a smooth pollution relation.

Figure 7 rules out a simple symmetric green-buying story. The effect comes from divesting brown stocks when pollution is salient. Figure 8 and Table 3 provide a high-frequency salience test: when AQI crosses 200, the government issues amber alerts, and treated investors discretely reduce brown share. Control PIN Codes do not show a similar discontinuity. This is particularly strong evidence for salience because the threshold has public communication meaning.

Investors respond most clearly when the information is attention-grabbing. The salient AQI warning threshold produces a discrete shift away from brown stocks.



(a) Figure 7: Selling brown versus buying green.



(b) Figure 8: RD plots around the AQI 200 warning threshold.

Trading direction and threshold salience.

intervals) around the critical AQI value of 200, which is the cutoff between “Moderate” and “Poor”.

Table 3

Regression discontinuity estimates: AQI transition from “Moderate” to “Poor”.

(1)	(2)	(3)	(4)	(5)	(6)
smoothness constant	M	band-width	estimate	Lower CI	Upper CI
Panel A Treated PIN Codes					
Kernel = Uniform					
0.1*M_rot	0.0001	41	-0.65	-0.8297	-0.4703
0.5*M_rot	0.0007	21	-0.9145	-1.1638	-0.6652
M_rot	0.0014	16	-0.8878	-1.1747	-0.6009
5*M_rot	0.0071	8	-0.6721	-1.0825	-0.2616
10*M_rot	0.0143	6	-0.9334	-1.4149	-0.4519
Kernel = Triangle					
0.1*M_rot	0.0001	52.36	-0.6813	-0.8565	-0.5061
0.5*M_rot	0.0007	27.85	-0.8361	-1.0793	-0.593
M_rot	0.0014	20.92	-0.8513	-1.1321	-0.5705
5*M_rot	0.0071	11.20	-0.7361	-1.1386	-0.3337
10*M_rot	0.0143	8.38	-0.7461	-1.2204	-0.2718
Panel B Control PIN Codes					
Kernel = Uniform					
0.1*M_rot	0.0001	40	-0.117	-0.2712	0.0372
0.5*M_rot	0.0007	21	-0.1156	-0.3298	0.0986
M_rot	0.0013	15	-0.0892	-0.3372	0.1588
5*M_rot	0.0067	8	-0.0001	-0.3511	0.3509
10*M_rot	0.0134	6	-0.167	-0.5776	0.2436
Kernel = Triangle					
0.1*M_rot	0.0001	50.64	-0.1372	-0.2874	0.013
0.5*M_rot	0.0007	26.94	-0.0812	-0.2898	0.1275
M_rot	0.0013	20.24	-0.0999	-0.3412	0.1413
5*M_rot	0.0067	10.83	-0.1169	-0.4601	0.2263
10*M_rot	0.0134	8.16	-0.1439	-0.5453	0.2575

Table 3: Regression discontinuity estimates at the Moderate-to-Poor AQI transition.

7.6 Figure 9: Heterogeneity by trading technology, age, and gender

Read:

- Panel A shows stronger effects for mobile traders, followed by internet traders, and weaker effects for traditional trading channels.

- Panel B shows stronger responses for young investors and weaker responses for older investors.
- Panel C shows stronger responses for women than men.
- The heterogeneity results are exploratory because these investor traits are not randomly assigned.

Figure 9 links the main effect to investor characteristics plausibly related to information access and environmental concern. Mobile traders are more likely to see AQI updates through phone-based channels; young investors may be more environmentally attentive; and female investors show a stronger response consistent with prior environmental-concern evidence. The main caution is that these differences cannot by themselves identify causal subgroup mechanisms.

The response to environmental information is not homogeneous. Information salience matters more for investors who are more likely to see the information or to care about it.

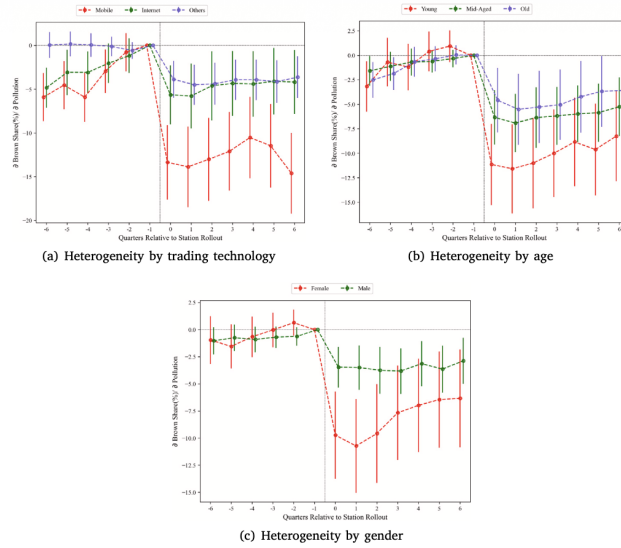


Fig. 9. Heterogeneity analyses of changes in brown-share-pollution gradient.
Note: This figure shows the sensitivity of brown investment to pollution before and after the installation of Continuous Ambient Air Quality Monitoring Stations for different groups of investors. Specifically we plot the coefficients $\{\beta_i\}$ from our preferred specification in Table 2, column 8, splitting the investor sample by various attributes. Panel A shows heterogeneity by trading technology, Panel B shows heterogeneity by age, and Panel C shows heterogeneity by gender. In each case, the graph shows the estimated coefficients $\{\beta_i\}$ as well as 95% confidence intervals obtained from standard errors clustered at the PIN Code level.

Figure 9: Heterogeneity analyses of changes in the brown-share-pollution gradient.

8 Result map

- **Result 1: Public pollution information matters.** Retail investors' brown share becomes more negatively related to local pollution after a nearby station appears.
- **Result 2: No pretrend in pollution or portfolio sensitivity.** Pollution itself does not jump around station rollout, and brown-share sensitivity is flat before treatment.
- **Result 3: Donut controls validate local relevance.** Control PIN Codes 40–60 km from the same station show no comparable change.
- **Result 4: The effect is brown divestment.** Order-imbalance evidence shows selling brown stocks drives the result, not green buying.
- **Result 5: Salience is strengthened by warnings.** The AQI 200 warning threshold produces a discontinuous reduction in treated investors' brown share.
- **Result 6: Effects are heterogeneous.** Mobile, young, and female investors respond more strongly.

- **Result 7: Possible return cost.** Brown stocks sold on high-pollution days subsequently earn positive excess returns, suggesting nonpecuniary motives or belief changes.

9 Short summary

- The paper studies whether making local pollution information salient changes retail investor portfolio allocations.
- It exploits the staggered rollout of CAAQMS pollution monitoring stations in India.
- Treated investors live within 20 km of a station; controls live in a 40–60 km donut ring.
- Retail portfolios are built from NSE trading records for about 19 million investors over 2004–2020.
- Brown share becomes more negatively related to local pollution after nearby monitoring information appears.
- The effect is absent before rollout and absent among donut-control investors.
- The shift is driven by brown-stock sales rather than green-stock purchases.
- A discontinuity at the AQI 200 public warning cutoff strengthens the salience interpretation.
- Effects are strongest for mobile traders, young investors, and female investors.
- The paper contributes evidence that information salience can change retail ESG behavior and may impose return costs.

10 Conclusion

10.1 Core conclusion

Fisman, Ghosh, Sarkar, and Zhang show that ready access to local pollution information changes how retail investors allocate portfolios between brown and green stocks. The appearance of nearby air-quality stations makes local pollution salient, and investors respond by reducing brown stock holdings when local pollution is high.

10.2 Economic intuition

Local pollution may be present before a monitoring station appears, but investors may not attend to it. A real-time station changes the informational environment: AQI readings become easy to observe and personally relevant. When pollution becomes salient, investors are more willing to act on environmental concerns in their portfolios.

10.3 Incremental contribution revisited

The paper contributes to sustainable finance by showing that retail ESG demand is not fixed; it can be activated by information salience. It contributes to household finance by using actual trading data rather than survey attitudes. It contributes to environmental economics by showing that pollution-information infrastructure changes financial behavior, not just health avoidance or search behavior.

10.4 Policy implication

If policymakers want household portfolios to tilt away from brown firms, information disclosure may matter, but only if it is salient, local, and easy to access. Generic ESG disclosure may be weaker than direct, real-time environmental information that investors can connect to lived experience.

10.5 Limitations

The paper cannot fully distinguish whether investors sell brown stocks because they update expected returns, because they experience stronger green tastes, or because pollution salience triggers warm-glow motives. Heterogeneity evidence is suggestive rather than causal. Institutional investors do not show comparable responses, so the results are most directly about retail investors.

10.6 Final takeaway

The central takeaway is that dirty air changes portfolios only when investors can see it. Information infrastructure turns environmental conditions into salient investment inputs, and that salience shifts retail investors away from brown stocks.